Department of Mathematics, IIT Madras
MA2101 Linear Algebra via Matrices
Assignment 6

---

1. Let $A = \begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix}$, $B = \begin{bmatrix} -1 & 3 \\ 0 & 0 \end{bmatrix}$ and $C = \begin{bmatrix} 0 & 3 \\ 0 & -1 \end{bmatrix}$. Show that both B and C are Schur triangularizations of A.
2. Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 \\ 0 & 7 \end{bmatrix}$ and $C = \begin{bmatrix} 0 & 2 \\ 0 & 7 \end{bmatrix}$. Show that B is a Schur triangularization of A but C is not.
3. Determine a matrix A such that $A^*A = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 1 & i \\ 0 & -i & 1 \end{bmatrix}$.
4. Use the formula $(A - tI)\,\mathrm{adj}(A - tI) = \det(A - tI)I$ to give another proof of Cayley-Hamilton theorem. [Hint: For matrices $B_0, \ldots, B_{n-1}$, write $\mathrm{adj}(A - tI) = B_0 + B_1 t + \cdots + B_{n-1}t^{n-1}$.]
5. Let $A \in \mathbb{F}^{n\times n}$. Let $p(t)$ be a monic polynomial of degree n. If $p(A) = 0$, then does it follow that $p(t)$ is the characteristic polynomial of A? NO
6. Diagonalize the given matrix, and then compute its fifth power:
   (a) $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 7 & -2 & 0 \\ -2 & 6 & -2 \\ 0 & -2 & 5 \end{bmatrix}$ (c) $\begin{bmatrix} 7 & -5 & 15 \\ 6 & -4 & 15 \\ 0 & 0 & 1 \end{bmatrix}$
7. Show that the following matrices are diagonalized by matrices in $\mathbb{R}^{3\times 3}$.
   (a) $\begin{bmatrix} 3/2 & -1/2 & 0 \\ -1/2 & 3/2 & 0 \\ 1/2 & -1/2 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 3 & -1/2 & -3/2 \\ 1 & 3/2 & 3/2 \\ -1 & 1/2 & 5/2 \end{bmatrix}$ (c) $\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & 0 \\ 2 & 2 & 3 \end{bmatrix}$
8. If possible, diagonalize the following matrices:
   (a) $\begin{bmatrix} 2 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 2 & -1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & -2 & 1 \\ 0 & 1 & 1 \\ 0 & 3 & -1 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ -1 & -2 & -3 \end{bmatrix}$
9. Are the following matrices diagonalizable?
   (a) $\begin{bmatrix} 2 & 3 \\ 6 & -1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & -10 & 0 \\ -1 & 3 & 1 \\ -1 & 0 & 4 \end{bmatrix}$ (c) $\begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 5 \end{bmatrix}$
10. Check whether each of the following matrices is diagonalizable. If diagonalizable, find a basis of eigenvectors for $\mathbb{C}^{3\times 1}$:
    (a) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$
11. Find orthogonal or unitary diagonalizing matrices for the following:
    (a) $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 3+i \\ 3-i & 4 \end{bmatrix}$ (c) $\begin{bmatrix} -4 & -2 & 2 \\ -2 & -1 & 1 \\ 2 & 1 & -1 \end{bmatrix}$

    hermitian ⇒ can be diagonalized by unitary matrix
12. Determine A^5 and a matrix B such that $B^2 = A$, where
    (a) $A = \begin{bmatrix} 2 & 1 \\ -2 & -1 \end{bmatrix}$ (b) $A = \begin{bmatrix} 9 & -5 & 3 \\ 0 & 4 & 3 \\ 0 & 0 & 1 \end{bmatrix}$
13. If A is a normal matrix, then show that A^*, $I + A$ and A^2 are also normal.

1. Let $A = \begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix}$, $B = \begin{bmatrix} -1 & 3 \\ 0 & 0 \end{bmatrix}$ and $C = \begin{bmatrix} 0 & 3 \\ 0 & -1 \end{bmatrix}$. Show that both B and C are Schur triangularizations of A.

$$\begin{vmatrix} \lambda-1 & -2 \\ 1 & \lambda+2 \end{vmatrix} = 0 \Rightarrow \lambda^2+\lambda-2+2=0 \Rightarrow \lambda = 0, -1$$

eigenvector corresponding to $\lambda=0$:

$$v_1+2v_2=0 \Rightarrow \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

make an orthonormal basis: $P = \begin{bmatrix} -2/\sqrt{5} & 1/\sqrt{5} \\ 1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix}$

$$P^*AP = \begin{bmatrix} -2/\sqrt{5} & 1/\sqrt{5} \\ 1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} -2/\sqrt{5} & 1/\sqrt{5} \\ 1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix}$$

$$= \begin{bmatrix} -3/\sqrt{5} & -6/\sqrt{5} \\ -1/\sqrt{5} & -2/\sqrt{5} \end{bmatrix} \begin{bmatrix} -2/\sqrt{5} & 1/\sqrt{5} \\ 1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -3 \\ 0 & -1 \end{bmatrix}$$

eigenvector corresponding to $\lambda=-1$

$$\begin{matrix} v_1+2v_2=-v_1 \\ -v_1-2v_2=-v_2 \end{matrix} \Rightarrow \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

make an orthonormal basis: $\begin{bmatrix} -1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} = P$

$$P^*AP = \begin{bmatrix} -1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} -1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

$$= \begin{bmatrix} -2/\sqrt{2} & -4/\sqrt{2} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} -1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} = \begin{bmatrix} -1 & -3 \\ 0 & 0 \end{bmatrix}$$

13. If A is a normal matrix, then show that A^*, $I + A$ and A^2 are also normal.

$AA^* = A^*A$

$(I+A)(I+A^*) = I + A + A^* + AA^* = I + A + A^* + A^*A = A^*(I+A) + (I+A)$

$= (I+A^*)(I+A)$

$A^2(A^2)^* = A \cdot A \cdot A^* A^* = AA^* AA^* = A^* AAA^* = A^* AA^* A = A^* A^* AA$

$= (A^*)^2 A^2$

$= (A^2)^* A^2$

6. Diagonalize the given matrix, and then compute its fifth power:

(a) $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 7 & -2 & 0 \\ -2 & 6 & -2 \\ 0 & -2 & 5 \end{bmatrix}$ (c) $\begin{bmatrix} 7 & -5 & 15 \\ 6 & -4 & 15 \\ 0 & 0 & 1 \end{bmatrix}$

(a) $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} \lambda & -1 & -1 \\ -1 & \lambda & -1 \\ -1 & -1 & \lambda \end{bmatrix} = \lambda[\lambda^2 - 1] + 1[-\lambda - 1]$

$+ (-1)[1 + \lambda]$

$= \lambda^3 - 3\lambda - 2 = 0$

$\lambda = -1, 2, -1$

$\lambda = -1$: $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$ $\lambda = 2$: $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

$\therefore$

Linearly independent:

$P = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ -1 & -1 & 1 \end{bmatrix}$ $D = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

$AP = PD$

$$A = PDP^{-1} \Rightarrow A^{5} = PD^{5}P^{-1}$$

$$= \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 32 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ -1 & -1 & 1 \end{bmatrix}^{-1}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ -1 & -1 & 1 & 0 & 0 & 1 \end{array}\right]$$

$$\downarrow$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & -1 & 2 & 1 & 1 & 1 \end{array}\right]$$

$$\downarrow$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & \frac{1}{3} & \frac{2}{3} & \frac{1}{3} \end{array}\right]$$

$$\downarrow$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 2/3 & -2/3 & -1/3 \\ 0 & 1 & 0 & -1/3 & 1/3 & -1/3 \\ 0 & 0 & 1 & 1/3 & 2/3 & 1/3 \end{array}\right]$$

$$\begin{bmatrix} -1 & 0 & 32 \\ 0 & -1 & 32 \\ 1 & 1 & 32 \end{bmatrix} \begin{bmatrix} 2/3 & -2/3 & -1/3 \\ -1/3 & 1/3 & -1/3 \\ 1/3 & 2/3 & 1/3 \end{bmatrix} = \begin{bmatrix} 10 & 11 & 11 \\ 11 & 10 & 11 \\ 11 & 11 & 10 \end{bmatrix}$$

4. Use the formula $(A - tI)\,\mathrm{adj}(A - tI) = \det(A - tI)I$ to give another proof of Cayley-Hamilton theorem. [Hint: For matrices $B_0, \ldots, B_{n-1}$, write $\mathrm{adj}(A - tI) = B_0 + B_1 t + \cdots + B_{n-1}t^{n-1}$.]

$$(A - tI)(B_0 + B_1 t + \cdots + B_{n-1}t^{n-1}) = p(t) \cdot I$$

$$t = A \Rightarrow 0 = p(A) \cdot I \Rightarrow p(A) = 0$$

7. Show that the following matrices are diagonalized by matrices in $\mathbb{R}^{3\times 3}$.

(a) $\begin{bmatrix} 3/2 & -1/2 & 0 \\ -1/2 & 3/2 & 0 \\ 1/2 & -1/2 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 3 & -1/2 & -3/2 \\ 1 & 3/2 & 3/2 \\ -1 & 1/2 & 5/2 \end{bmatrix}$ (c) $\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & 0 \\ 2 & 2 & 3 \end{bmatrix}$

$$\begin{bmatrix} \lambda - 3/2 & 1/2 & 0 \\ 1/2 & \lambda - 3/2 & 0 \\ \cdot & \cdot & \lambda - 1 \end{bmatrix} \Rightarrow (\lambda - 1)(\lambda^2 + 9/4 - 3\lambda - 1/4)$$

$$\Rightarrow \lambda = 1, 1, 2$$

$$3/2\,\theta_1 - 1/2\,\theta_2 = \theta_1 \quad -1/2\,\theta_1 + 3/2\,\theta_2 = \theta_2 \Rightarrow \theta_1 = \theta_2$$

$$\theta_1/2 - \theta_2/2 + \theta_3 = \theta_3 \Rightarrow \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\theta_1 = -\theta_2 = \theta_3 \Rightarrow \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

12. Determine A^5 and a matrix B such that $B^2 = A$, where

(a) $A = \begin{bmatrix} 2 & 1 \\ -2 & -1 \end{bmatrix}$ (b) $A = \begin{bmatrix} 9 & -5 & 3 \\ 0 & 4 & 3 \\ 0 & 0 & 1 \end{bmatrix}$

$$A = \begin{bmatrix} 9 & -5 & 3 \\ 0 & 4 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

$\lambda = 1, 4, 9$

$9\theta_1 - 5\theta_2 + 3\theta_3 = \theta_1$
$4\theta_2 + 3\theta_3 = \theta_2 \Rightarrow \theta_2 = -\theta_3 = \theta_1$
$\theta_3 = \theta_3$

$\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$

$9\theta_1 - 5\theta_2 + 3\theta_3 = 4\theta_1$
$4\theta_2 + 3\theta_3 = 4\theta_2 \Rightarrow \theta_3 = 0 \Rightarrow \theta_1 = \theta_2$
$\theta_3 = 4\theta_3$

$\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

$9\theta_1 - 5\theta_2 + 3\theta_3 = 9\theta_1$
$4\theta_2 + 3\theta_3 = 9\theta_2 \quad \theta_3 = 0, \theta_2 = 0$
$\theta_3 = 9\theta_3$

$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

$$P = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix} \qquad D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

$$A^5 = PD^5P^{-1} = \begin{bmatrix} 3^5 & 2^{10} - 3^5 & 2^{10} - 1 \\ 0 & 2^{10} & 2^{10} - 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$B = PD^{1/2}P^{-1} = \begin{bmatrix} 3 & -1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

14. Determine the Jordan forms of the following matrices:

(a) $\begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 2 & 1 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

15. Let A be an $n \times n$ a lower triangular matrix with each diagonal entry as 1, each sub-diagonal entry (just below each diagonal entry) as 1, and all other entries being 0. What is the Jordan form of A?

16. Let A be an $n \times n$ matrix, where the diagonal entries are $1, 2, \ldots, n$, from top left to right bottom, the super diagonal entries are all 1, and all other entries are 0. What is the Jordan form of A?

17. What is the Jordan form of the $n \times n$ matrix whose each row is equal to $[1, 2, \cdots, n]$?

18. Let $A \in \mathbb{C}^{m \times n}$. Show that the positive singular values of A are also the positive singular values of A^*.

19. Compute the singular value decompositions of the following matrices:

(a) $\begin{bmatrix} 2 & -2 \\ 1 & -1 \\ -1 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 2 & 1 & 2 \\ -2 & 1 & 2 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 2 & 2 \\ 2 & 0 & -5 \\ 3 & 0 & 0 \end{bmatrix}$

20. Show that the matrices $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ and $\begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix}$ are similar but they have different singular values.

21. Prove that if $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of an $n \times n$ hermitian matrix, then its singular values are $|\lambda_1|, \ldots, |\lambda_n|$.

22. Let $A \in \mathbb{C}^{n \times n}$ and let $\lambda \in \mathbb{C}$. Show that $A - \lambda I$ is invertible iff $A^* - \overline{\lambda} I$ is invertible.

23. Let $A \in \mathbb{F}^{n \times n}$. If A has eigenvalues $\lambda_1, \ldots, \lambda_n$ and singular values $s_1, \ldots, s_n$, then show that $|\lambda_1 \cdots \lambda_n| = s_1 \cdots s_n$.

$$A^*A = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 1 & i \\ 0 & -i & 1 \end{bmatrix}$$

$A \in C^{n\times n} \rightarrow$ normal $\Rightarrow A^*A = AA^*$

$\Updownarrow$

$A = UDU^*$ for some $U \in C^{n\times n}$ unitary

$D \in C^{n\times n}$ diagonal $\rightarrow \sigma(A) = \sigma(D)$

A is said to be positive semidefinite if $A = B^*B$ for some $B \in C^{n\times n}$

A is positive semidefinite $\Leftrightarrow$ A is normal $\Leftrightarrow$ A is hermitian

$\sigma(A) \in [0, \infty)$

$$D = \begin{bmatrix} d_1 & & & \\ & d_2 & & \\ & & \ddots & \\ & & & d_n \end{bmatrix} \Rightarrow A^{1/2} = UD^{1/2}U^* \rightarrow \text{normal} + \sigma(A^{1/2}) = \{\sqrt{d_1} \dots \sqrt{d_n}\}$$

$\Rightarrow A^{1/2}$ is +ve semidefinite

$$A = \left[\begin{array}{c|cc} \lambda & 0 & \\ \hline 0 & & B_{2\times 2} \end{array}\right] \Rightarrow A^*A = \left[\begin{array}{c|cc} \lambda^2 & 0 & 0 \\ \hline 0 & & \\ 0 & & B^*B_{2\times 2} \end{array}\right] \qquad \boxed{\lambda = 2}$$

$$A_1 = B^*B = \begin{bmatrix} 1 & i \\ -i & 1 \end{bmatrix} \Rightarrow \sigma(A_1) \subseteq [0, \infty),\ A_1 \text{ is hermitian}$$

$\Rightarrow A_1$ is +ve semidefinite

take $B = (A_1)^{1/2}$

$A_1 = UDU^*$ $\qquad D = \begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix}$; $U = \begin{bmatrix} 1/\sqrt{2} & i/\sqrt{2} \\ i/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$

$$B=\begin{bmatrix}1/\sqrt{2} & i/\sqrt{2}\\ i/\sqrt{2} & 1/\sqrt{2}\end{bmatrix}\begin{bmatrix}0 & 0\\ 0 & \sqrt{2}\end{bmatrix}\begin{bmatrix}1/\sqrt{2} & -i/\sqrt{2}\\ -i/\sqrt{2} & 1/\sqrt{2}\end{bmatrix}$$

$$=\begin{bmatrix}0 & i\\ 0 & 1\end{bmatrix}\begin{bmatrix}1/\sqrt{2} & -i/\sqrt{2}\\ -i/\sqrt{2} & 1/\sqrt{2}\end{bmatrix}=\begin{bmatrix}1/\sqrt{2} & i/\sqrt{2}\\ -i/\sqrt{2} & 1/\sqrt{2}\end{bmatrix}$$

$$A=\begin{bmatrix}2 & 0 & 0\\ 0 & 1/\sqrt{2} & i/\sqrt{2}\\ 0 & -i/\sqrt{2} & 1/\sqrt{2}\end{bmatrix}$$

(a) $\begin{bmatrix}2 & -2\\ 1 & -1\\ -1 & 1\end{bmatrix}$ rank = 1

$$A^*A=\begin{bmatrix}2 & 1 & -1\\ -2 & -1 & 1\end{bmatrix}\begin{bmatrix}2 & -2\\ 1 & -1\\ -1 & 1\end{bmatrix}=\begin{bmatrix}6 & -6\\ -6 & 6\end{bmatrix}$$

$\lambda=\{0,12\}$

$S_1=2\sqrt{3} \Rightarrow \Sigma=\begin{bmatrix}2\sqrt{3} & 0\\ 0 & 0\\ 0 & 0\end{bmatrix}$

$\Rightarrow AA^*u$